Topics in Valuation

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Estimating TVs: We Have a Problem

- We think about companies as potentially having an infinite life. Therefore, value should be the PV of pay-offs forever:

$$V_0 = \sum_{t=1}^{\infty} \frac{X_t}{(1+r)^t}$$

- Since we cannot estimate cash flows forever, we estimate cash flows for a "forecasting period" and then estimate a terminal value to capture the value beyond the forecasting period

$$V_0 = \sum_{t=1}^{n} \frac{X_t}{(1+r)^t} + \frac{TV_n}{(1+r)^n}$$

- This is just a math trick (best we can do) but, implicitly requires a firm to be in **steady-state** by then

Steady-state perpetual growth

- Never can exceed the growth rate of the economy but it can be lower
- There will always be growth and mature (steady-state) companies in the economy, steady-state firms will have to grow at a lower rate than the economy
- Negative perpetual growth? Yes, it's possible. Declining businesses: Tobacco? Fossil fuel producing Vs Clean energy? etc...
- Nominal or real growth rates?
- Proxy for long-term growth of the economy? Long-run GDP average growth rate reasonable one

Top Down Fundamental Forecasts

- Never add detail until you can reasonably think that adding details can help building-up your story
- Common approach: 3 main drivers
  1. Estimate growth rates in revenue over time: total market revenue and % your target company will earn
  2. Set a target margin for steady-state that aligns with your expected revenue growth rate (OI to Sales)
  3. Estimate the capital needed to sustain your expected revenue growth (Sales to capital)
- A RNOA forecast encompasses these three at once (Obviously: RNOA = OI/Sales x Sales/NOA)
- The steady-state condition is unavoidable (more granular details are not a way around it)

Top Down Forecasts: Example

Years	Sales	Sales g (1)	Operating Margin (2)	Sales to Capital (3)
0	€ 3,625,731		-13.69%	2
1	€ 5,076,023	40%	-10%	2
2	€ 6,345,029	25%	-7.56%	2
3	€ 7,931,287	25%	-5.11%	2
4	€ 9,914,108	25%	-2.67%	2
5	€ 12,392,635	25%	-0.22%	2
6	€ 14,920,733	20.40%	2.22%	2
7	€ 17,278,209	15.80%	4.67%	2
8	€ 19,213,368	11.20%	7.11%	2
9	€ 20,481,450	6.60%	9.56%	2
10	€ 20,891,079	2%	12%	2
Terminal	€ 21,308,901	2%	12%	2

Top Down Forecasts: Example

Years	Sales	Sales g (1)	Operating Margin (2)	Sales to Capital (3)	OI	OI (1-t)	beg NOA	RNOA
0	€ 3,625,731		-13.69%	2	-€ 496,363	-€ 496,363	€ 1,812,866	-27.4%
1	€ 5,076,023	40%	-10%	2	-€ 507,602	-€ 507,602	€ 2,538,012	-20.0%
2	€ 6,345,029	25%	-7.56%	2	-€ 479,402	-€ 479,402	€ 3,172,515	-15.1%
3	€ 7,931,287	25%	-5.11%	2	-€ 405,377	-€ 405,377	€ 3,965,643	-10.2%
4	€ 9,914,108	25%	-2.67%	2	-€ 264,376	-€ 264,376	€ 4,957,054	-5.3%
5	€ 12,392,635	25%	-0.22%	2	-€ 27,539	-€ 27,539	€ 6,196,318	-0.4%
6	€ 14,920,733	20.40%	2.22%	2	€ 331,572	€ 331,572	€ 7,460,366	4.4%
7	€ 17,278,209	15.80%	4.67%	2	€ 806,316	€ 806,316	€ 8,639,104	9.3%
8	€ 19,213,368	11.20%	7.11%	2	€ 1,366,284	€ 614,907	€ 9,606,684	6.4%
9	€ 20,481,450	6.60%	9.56%	2	€ 1,957,116	€ 1,467,837	€ 10,240,725	14.3%
10	€ 20,891,079	2%	12%	2	€ 2,506,930	€ 1,880,197	€ 10,445,540	18.0%
Terminal	€ 21,308,901	2%	12%	2	€ 2,557,068	€ 1,917,801	€ 10,654,450	18.0%

Top Down Forecasts: Example

Years	Sales	Sales g (1)	Operating Margin (2)	Sales to Capital (3)	OI	OI (1-t)	beg NOA	RNOA
0	€ 3,625,731		-13.69%	2	-€ 496,363	-€ 496,363	€ 1,812,866	-27.4%
1	€ 5,07[illegible]	[illegible]	[illegible]	2	-€ 507,602	-€ 507,602	€ 2,538,012	-20.0%
2	€ 6,34[illegible]			2	-€ 479,402	-€ 479,402	€ 3,172,515	-15.1%
3	€ [illegible]7,93[illegible]			2	-€ 405,377	-€ 405,377	€ 3,965,643	-10.2%
4	€ 9,91[illegible]			2	-€ 264,376	-€ 264,376	€ 4,957,054	-5.3%
5	€ 12,39[illegible]			2	-€ 27,539	-€ 27,539	€ 6,196,318	-0.4%
6	€ 14,92[illegible]			2	€ 331,572	€ 331,572	€ 7,460,366	4.4%
7	€ 17,27[illegible]			2	€ 806,316	€ 806,316	€ 8,639,104	9.3%
8	€ 19,21[illegible]			2	€ 1,366,284	€ 614,907	€ 9,606,684	6.4%
9	€ 20,48[illegible]			2	€ 1,957,116	€ 1,467,837	€ 10,240,725	14.3%
10	€ 20,8[illegible]			2	€ 2,506,930	€ 1,880,197	€ 10,445,540	18.0%
Terminal	€ 21,308,901	2%	12%	2	€ 2,557,068	€ 1,917,801	€ 10,654,458	18.0%

Sanity Check:
Too high (low) implying reinvestment needed to sustain revenue growth is probably too low (high)

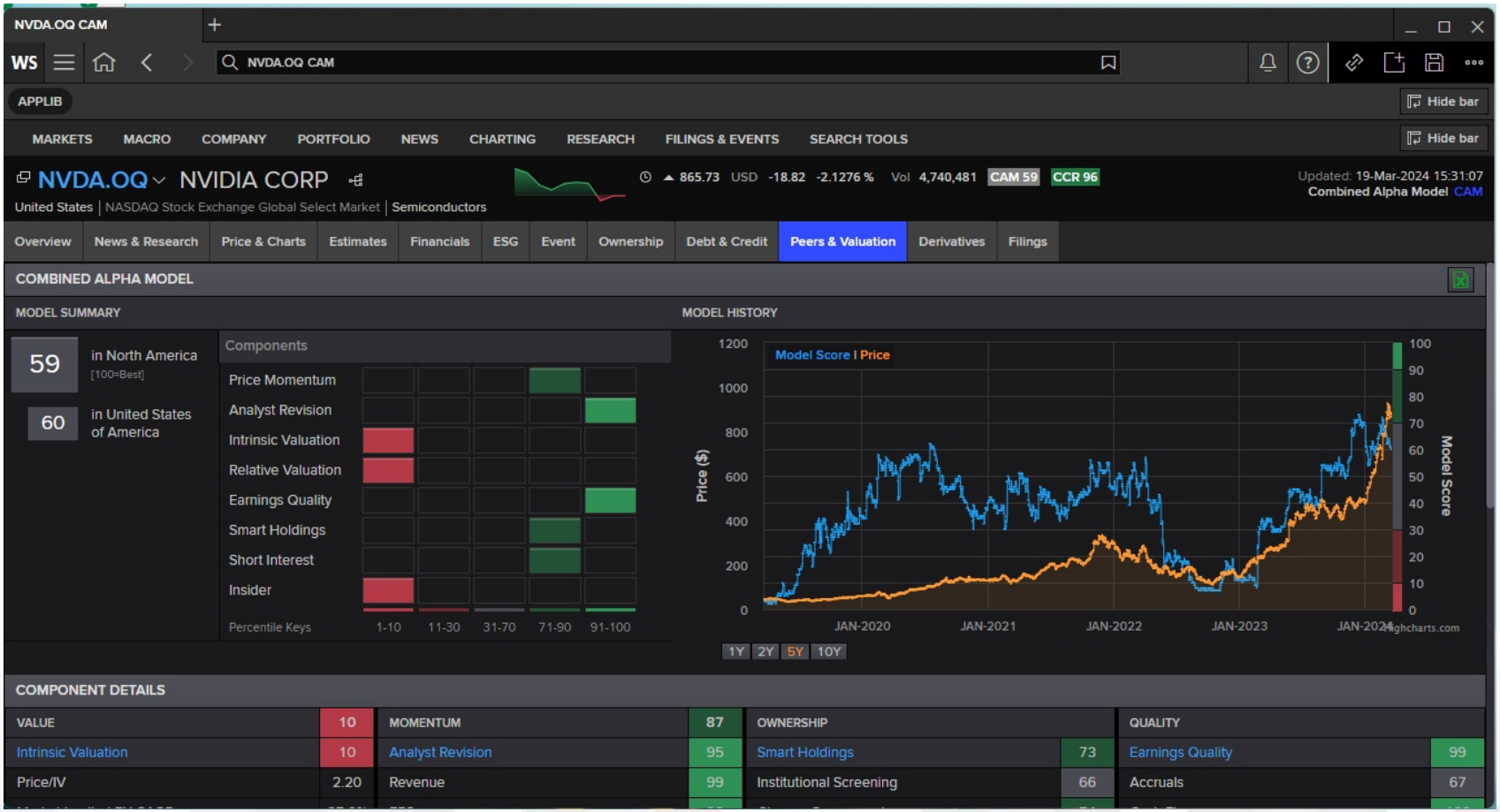
Top Down Forecasts: Example

- What is the implied FCF on the previous forecast?

Top Down Forecasts: Example

Years	OI	OI (1-t)	beg NOA	RNOA	FCF
0	-€ 496,363	-€ 496,363	€ 1,812,866	-27.4%	
1	-€ 507,602	-€ 507,602	€ 2,538,012	-20.0%	-€ 1,232,749
2	-€ 479,402	-€ 479,402	€ 3,172,515	-15.1%	-€ 1,113,905
3	-€ 405,377	-€ 405,377	€ 3,965,643	-10.2%	-€ 1,198,506
4	-€ 264,376	-€ 264,376	€ 4,957,054	-5.3%	-€ 1,255,787
5	-€ 27,539	-€ 27,539	€ 6,196,318	-0.4%	-€ 1,266,803
6	€ 331,572	€ 331,572	€ 7,460,366	4.4%	-€ 932,477
7	€ 806,316	€ 806,316	€ 8,639,104	9.3%	-€ 372,421
8	€ 1,366,284	€ 614,907	€ 9,606,684	6.4%	€ 398,704
9	€ 1,957,116	€ 1,467,837	€ 10,240,725	14.3%	€ 1,323,075
10	€ 2,506,930	€ 1,880,197	€ 10,445,540	18.0%	€ 2,302,115
Terminal	€ 2,557,068	€ 1,917,801	€ 10,654,450	18.0%	€ 2,348,157

Top Down Forecasts: Example

- What is the implied FCF on the previous forecast?
- 7 years of negative FCFs, could have FCF been positive sooner? How?
- Top down forecasts may end up resulting in excessively high/low RNOAs:
- If too high (low), is probably implying reinvestment needed to sustain growth is probably not enough (too much)
- Good sanity check (feedback): starting with RNOA avoids unnecessary feedback loops

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